INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL PRACTICAL & LABORATORY MANUAL â¢ DEPARTMENT OF CSE

Laboratory Course: [MCS-151] Colloquium & Research Review Paper-I

Programme: Master of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Dr. A. K. Ganguly (Head of D**

Official Laboratory Manual: Architecture Simulation & Cache Profiling

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1. Advanced Architecture Simulation Goals

- Platform: Gem5 cycle-accurate simulator / SimpleScalar toolkit on Linux.
- Objective: Empirical evaluation of speculative branch prediction, cache miss rates, and multi-core IPC.
- Grading: Simulation trace analysis, IPC metrics graphs, and research lab report.

2. Experimental Assignments

- Experiment 1-2: Gem5 configuration for Out-of-Order (O3) CPU model and IPC benchmark comparisons.
- Experiment 3-4: L1/L2 cache associativity variation and miss rate characterization.
- Experiment 5: Shared memory multi-core coherence trace under MESI protocol.